

Homework 02

CS 355 Database Systems
Habib University
Fall 2025

1 Instructions

- The deadline for submitting this Homework is **November 27th, 2025, 11:59 PM**.
- **No extension will be granted for this homework, so start early to be able to finish on time. Any requests related to extension will result in 0 grade for your homework directly.**
- Submit your solution as a single PDF file via **Canvas**.
- Name your file as: **HW_02_team_XX.pdf**, where XX is your Canvas team number.
- This homework must be completed in pairs.
- Clearly state any assumptions you make. They should not contradict the question.
- Use of AI tools such as ChatGPT is strictly prohibited and will be considered plagiarism.

1.1 Marking Scheme

- Question 1: [20 marks]
- Question 2: [20 marks]
- Question 3: [20 marks]
- Question 4: [20 marks]
- Question 5: [20 marks]

1.2 Late Submission Policy

Refer to the course syllabus for the late submission policy.

Question 1: Relational Algebra

Consider the following relational schema:

- Students(s_id, s_name, s_dept)
 - Courses(c_id, c_title, c_dept, c_credits)
 - Prerequisites(c_id, p_id)
 - Enrollments(c_id, s_id, e_grade, e_grade_point)
- (a) [4 points] Write a relational algebra expression to find out the names of students that have taken all courses from their department. **Output: s_name**
- (b) [4 points] Write a relational algebra expression that returns the IDs and names of students that have gotten a better grade in a course than in at least one of the prerequisites of the course. **Output: s_id, s_name**
- (c) [4 points] Write a relational algebra expression that returns the failed credits of each student. **Output: s_id, c_id, c_credits**
- (d) [4 points] Write a relational algebra expression that returns the titles of courses that have more than one prerequisite. **Output: c_title**
- (e) [4 points] Write a relational algebra expression that returns the c_id of courses that either the student with s_id 001 has taken or the student with s_id 002 has taken, but not both. **Output: c_id**

Question 2: Conflict Serializable/Cascadeless Schedules

Consider the following three transactions:

T1	T2	T3
Read(A)	Read(A)	Read(T)
Read(S)	Read(B)	Read(A)
Read(T)	$A = B + 1$	$T = A + T$
$A = S + T$	Write(A)	Write(T)
Write(A)	Commit	Commit
Commit		

- (a) [4 points] Provide a concurrent schedule of T1, T2, and T3 that is **not conflict-serializable**. Justify by highlighting the instructions that make it non-serializable in your solution.
- (b) [4 points] Provide a concurrent schedule of T1, T2, and T3 that is **conflict-serializable but not recoverable**. Justify by highlighting instructions that make it non-recoverable.
- (c) [4 points] Provide a concurrent schedule of T1, T2, and T3 that is **conflict-serializable and recoverable, but not cascadeless**. Justify by highlighting instructions that make it non-cascadeless in your solution.
- (d) [4 points] Provide a concurrent schedule of T1, T2, and T3 that is **conflict-serializable, recoverable, and cascadeless**.
- (e) [4 points] Provide a concurrent schedule of T1, T2, and T3 that is **conflict-serializable and cascadeless, but not recoverable**.

Question 3: Precedence Graph

Consider the following schedule:

- **S1:** r1(A), w1(A), r2(A), w1(C), r4(D), w2(B), r3(B), r1(A), r3(C), r4(C), w4(A), r2(B), w3(D)

- [10 points] Draw the serializability (precedence) graph for S1.
- [5 points] State whether the schedule is serializable or not, based on the graph.
- [5 points] If serializable, derive the equivalent serial schedule.

Question 4: Two-Phase Locking (2PL)

Consider a database system that employs the Two-Phase Locking (2PL) protocol for concurrency control. The system has transactions T1, T2, and T3. Assume the database has three items (A, B, C). The following tables show the sequence of read and write operations on these items.

- [10 points] For each transaction, identify where shared locks (Lock-S) or exclusive locks (Lock-X) should be applied.
- [10 points] Provide one possible conflict equivalent serializable schedule for the given transactions along with locks.

T1	T2	T3
Read(A)	Read(B)	Read(A)
Read(B)	Read(C)	Write(C)
Write(A)	Write(B)	Read(A)
Write(C)	Write(C)	Commit
Write(B)	Commit	
Read(C)		
Commit		

Question 5: NoSQL

Write MongoDB queries for the following tasks, using records inserted in a MongoDB collection called **"library"**, as shown in Listing 1.

- [2 points] Return the titles and authors of all the books in the database.
- [2 points] Return all books that were published between 1900 and 1999.
- [2 points] Return all the books that are not checked out by any member.
- [2 points] Display all titles and authors of books having **Fantasy** genre.
- [2 points] Insert a book with the following details:
 - Title: **"The Mortal Instruments - City of Bones"**
 - Authors: **Cassandra Clare, Joshua Lewis**
 - Genre: **Fantasy**
 - Publication Year: **2007**
 - Number of Copies Available: **5**

- (f) [2 points] Get the total number of books of the Horror genre.
- (g) [2 points] Display all the books that have more than one author.
- (h) [2 points] Increment the number of copies available of all the Fiction books by 3.
- (i) [2 points] Update all genres of “Fiction” to “Novels”.
- (j) [2 points] Delete the first non-fiction book in the list.

```
{
  "title": "The Great Gatsby",
  "author": ["F. Scott Fitzgerald"],
  "genre": "Fiction",
  "publicationYear": 1925,
  "copiesAvailable": 5,
  "checkedOutBy": [
    { "memberId": "1001", "dueDate": ISODate("2023-12-15") },
    { "memberId": "1002", "dueDate": ISODate("2023-11-30") }
  ]
},
{
  "title": "Good Omens",
  "author": ["Terry Pratchett", "Neil Gaiman"],
  "genre": "Horror",
  "publicationYear": 1990,
  "copiesAvailable": 2,
  "checkedOutBy": [
    { "memberId": "1001", "dueDate": ISODate("2023-12-15") },
    { "memberId": "1002", "dueDate": ISODate("2023-11-30") }
  ]
},
{
  "title": "The Talisman",
  "author": ["Stephen King", "Peter Straub"],
  "genre": "Fantasy",
  "publicationYear": 1984,
  "copiesAvailable": 1,
  "checkedOutBy": [
    { "memberId": "1007", "dueDate": ISODate("2023-12-18") }
  ]
},
{
  "title": "Neverwhere",
  "author": ["Neil Gaiman", "Lenny Henry"],
  "genre": "Fantasy",
  "publicationYear": 1996
}
```

Listing 1: MongoDB Records for the **"library"** collection